

PHILIP FILADELPHIA TOMASUI

COMPUTER SCIENCE STUDENT

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SUMMARY

An Informatics student at Udayana University who bridges the gap between robust back-end systems and intelligent data solutions. Experienced in developing web platforms and faculty inventory systems from scratch, ensuring secure database architecture and seamless user experiences. Beyond core web development, I build practical AI applications—ranging from natural language processing tools for cultural preservation to machine learning models for predictive analysis. As an active coordinator at the Student Innovation Centre, I leverage my technical and leadership skills to validate student tech projects and drive innovation.

EXPERIENCE

Student Innovation Centre (SIC), Universitas Udayana January, 2025 - Present
Officer – Talent and Competition Division

- Ran and helped deliver several SIC technology-focused work programs.
- Served as Manager of the Smart City Division, coordinating an intensive mentoring and project-validation program for Universitas Udayana's Gemastik delegation team.

Best Proposal Award — PKM-KC June 2025
PIONEER PKM 2025, Universitas Udayana

- Received the Best Proposal Award in the PKM-KC (Student Creativity Program – Innovative Work) category at PIONEER PKM 2025, organized by Universitas Udayana.
- Developed and presented an innovative technology-based proposal under the national student creativity initiative theme, competing among proposals submitted campus-wide.

EDUCATION

B.S. in Informatics, Universitas Udayana — GPA: 3.97/4.00
2024 - Present

PROJECTS

Student Innovation Centre (SIC) Website January 2025 – June 2025
Back-end Developer

- Designed a relational database architecture using MySQL and applied normalization techniques to ensure data integrity and query efficiency.
- Built a fully functional RESTful API with Express.js, including complete CRUD features for managing the organization's internal data.

Final Project – Rekayasa Perangkat Lunak November 2025 – January 2026
Back-end Developer

- Designed a relational database architecture using MySQL, applying normalization techniques to ensure data integrity and optimize query performance.
- Built fully functional CRUD features using Laravel's Eloquent ORM to manage the organization's internal data.

Tempe Iris Project (FMIPA Inventory Loan System)

May 2026 – June 2026

Back-end Developer

- Analyzed and prepared system documentation, including SRS, DFD, and ERD, to ensure a well-structured database architecture.
- Developed a faculty inventory management system with Laravel and MySQL, focused on efficient loan and return logic.
- Integrated automatic email notifications via Mailtrap to deliver real-time loan confirmation status to users.
- Implemented automatic document generation with Dompdf to produce dynamic, ready-to-download PDF loan receipts.

Orinimo Store

March 2026 - August 2026

Full-Stack Web Development

- Developed end-to-end CRUD functionality using Laravel's MVC architecture, including form request validation, Eloquent ORM for database operations, and dynamic Blade views for data management.
- Implemented a secure authentication system covering user login, registration, and forgot-password flow with token-based password reset, built on Laravel's built-in authentication features.
- Integrated the Fonte WhatsApp service to automate OTP verification and real-time notification delivery to users.
- Built a report generation module supporting PDF and Excel export for invoices and data recaps, streamlining reporting workflows.

PsychoLens AI: Emotion Analysis with Deep Learning (IndoBERT)

May 2026

Team Leader & AI Engineer

- Built an intelligent NLP system to map audience emotions and responses from social-media audio/video transcripts.
- Implemented a state-of-the-art IndoBERT Transformer model to capture contextual relationships in Indonesian text and classify them into 8 psychological emotion classes.

Traditional Balinese Gamelan Music Classification System

June 2026

Machine Learning Engineer & Data Scientist

- Built an automatic audio-based classification system using a Gamelan Music Dataset to categorize traditional Balinese instruments (Saron Barong, Demung, Peking, Bonang, Slenthem, Kendhang).
- Designed a comparative performance experiment using Machine Learning approaches (KNN and SVM) and Deep Learning (LSTM).
- Conducted comparative model evaluation and optimized the system architecture to reach a final classification accuracy of 96%.

Case-Based Reasoning (CBR) System for Learning-Style Classification

May 2026

AI Engineer

- Designed and built an AI-based expert system applying the Case-Based Reasoning (CBR) method.

Data-Driven Solution for Predicting MSME (UMKM) Success Using Hybrid Sampling (National data science competition — Data Science Informatics Festival)

June 2026

Team Leader & Data Scientist

- Designed a data-driven solution to predict the success and sustainability of MSMEs (UMKM) in order to reduce business failure rates, for a data-science competition.
- Designed a comparative engineering pipeline across 4 main experimental scenarios, testing three robust algorithms: AdaBoost, Random Forest, and Support Vector Machine (SVM).

Waste Image Classification

July 2026

(National data science competition: SATRIA DATA Big Data Challenge)

Team Leader & Data Scientist

- Designed an automated image processing pipeline to remove corrupted files, exact duplicates (MD5), and near-duplicates using perceptual hashing (pHash).
- Evaluated pretrained CNN architectures (MobileNet vs. EfficientNetB3) for 3-class waste classification, achieving a peak 0.972 validation Macro F1-score with fine-tuned EfficientNetB3.

Water Level Time Series Forecasting

July 2026

(National data science competition: Sebelas Maret Statistics Data Science)

Data Scientist

- Handled irregular timestamps, resolved missing values, and filtered sensor anomalies or outliers from historical hydrological time series data.
- Engineered lagged features, rolling statistical windows (moving averages, variance), and cyclical temporal attributes to capture seasonal trends and sequential water level dynamics.

PUBLICATIONS

Time-Series Electricity Consumption Prediction using SARIMA, LightGBM, and LSTM

August 2026

Lead Researcher / Author

Target Publication: Jurnal Nasional Teknologi Informasi dan Aplikasinya (JNATIA) — (Status: Submitted / Under Review)

- Preprocessed and structured historical electricity consumption data, handling missing entries, seasonality, and feature engineering for time-lagged predictors.
- Developed and benchmarked hybrid forecasting models spanning classical statistics (SARIMA), gradient boosting (LightGBM), and deep neural networks (LSTM).
- Evaluated predictive accuracy using metrics like RMSE and MAE to determine the optimal architecture for capturing complex power usage patterns.

SKILLS

- Programming Languages: PHP, JavaScript, Python, Java, C.
- Backend Development: Laravel (Eloquent ORM), Node.js, Express.js.
- Frontend & Web: React, HTML5/CSS3, TailwindCSS.
- Database & Systems: MySQL, Docker, CasaOS.
- Data Science & ML: ML pipelines, Deep Learning, ensemble models (AdaBoost, XGBoost, LightGBM).
- Tools & OS: Git, GitHub, Postman, Linux (Ubuntu Desktop).